Notas de aula - MAT01009

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I Introduction to Differential Equations

1 Definitions and terminology

Definição 1. An equation containing the derivatives of one or more unknown functions (or dependent variables) with respect to one or more independent variables is called a *differential equation* (*DE*).

If a differential equations contains only ordinary derivatives of one or more unknown functions with respect to a single independent variable, then the equation is called an *ordinary* differential equation (*ODE*). If a differential equation contains partial derivatives of one or more unknown functions with respect to two or more independent variables, then the equation is called a *partial* differential equation (*PDE*).

The *order* of a differential equation is the order of the highest derivative that appears in the equation. A *first-order* differential equation contains only first derivatives, a *second-order* differential equation contains second derivatives, and so on. We can express an nth-order ordinary differential equation by the general form

$$F(x, y, y', y'', \ldots, y^{(n)}) = 0 \tag{1}$$

where F is a real-valued function of $n + 2$ variables: x, y, y', y'', ..., $y^{(n)}$. We'll assume it's possible to solve for the highest-order derivative, so we can write the equation in the *normal* form

$$\frac{d^n y}{dx^n} = f(x, y, y', y'', \ldots, y^{(n-1)}). \tag{2}$$

where f is a real-valued continuous function.

An nth-order ordinary differential equation is said to be *linear* if F is linear in y, y', y'', ..., $y^{(n)}$. That is, if the equation can be written in the form

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \tag{3}$$

where $a_n(x), a_{n-1}(x), \ldots, a_0(x)$ and $g(x)$ are functions of x alone. If the equation is not linear, then it's called *nonlinear*, that is, if nonlinear functions of the dependent variable y or its derivatives, such as $\sin(y)$ or e^y, appear in the equation.

Definição 2. Any function ϕ, defined on an interval I and possessing at least n derivatives that are continuous on I, which when substituted into the equation reduces it to an identity, is called a *solution* of the differential equation on I.

The interval I is called the *interval of definition*, or *interval of existence*, or *interval of validity*, or *domain of the solution* ϕ.

2 Initial-value problems

Teorema 1. *(existence of a unique solution). Let R be a rectangular region in the xy-plane defined by $a \leq x \leq b$, $c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f/\partial y$ are continuous on R, then there exists some interval $I_0 : (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$ defined on I_0, that is a solution of the initial-value problem.*

II First-Order Differential Equations

1 Solution curves without a solution

2 Separable equations

Definição 1. A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y) \tag{4}$$

is called a *separable* first-order equation or to have *separable variables.*

3 Linear equations

Definição 2. A first-order differential equation of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \tag{5}$$

is called a *linear* first-order equation in y.

Solving a linear first-order equation

(i) put a linear first-order equation in standard form

$$\frac{dy}{dx} + P(x)y = f(x),$$

(ii) find the integrating factor $\mu(x) = e^{\int P(x)dx}$. there's no need to use a constant of integration here,

(iii) multiply the equation by $\mu(x)$ to obtain $\frac{d}{dx}[\mu(x)y] = \mu(x)f(x)$,

(iv) integrate both sides to obtain $\mu(x)y = \int \mu(x)f(x)dx + C$,

4 Exact equations

Definição 3. A differential expression $M(x,y)dx + N(x,y)dy$ is said to be an *exact differential* in a region R if there exists a function $\psi(x,y)$ such that $d\psi = M(x,y)dx + N(x,y)dy$ for every point (x,y) in R. A first-order differential equation

$$M(x,y)dx + N(x,y)dy = 0$$

is called an *exact equation* if the expression $M(x,y)dx + N(x,y)dy$ is an exact differential.

Teorema 1 (criterion for exact differential). *Let $M(x,y)$ and $N(x,y)$ be continuous and have continuous partial derivatives on a rectangle R in the xy-plane. Then a necessary and sufficient condition that $M(x,y)dx + N(x,y)dy$ be an exact differential is*

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \tag{6}$$

5 Solutions by substitutions

III Higher-Order Differential Equations

1 Theory of linear equations

Teorema 1 (existence of a unique solution). *Let $a_n(x), a_{n-1}(x), \ldots, a_0(x)$ and $g(x)$ be continuous on an interval I and let $a_n(x) \neq 0$ for every x in I. If $x = x_0$ is any point in I, then a solution $y(x)$ of the initial-value problem exists on I and is unique.*

Teorema 2 (superposition principle—homogeneous equations). *Let $y_1(x), y_2, \ldots, y_k(x)$ be solutions of a homogeneous n-th order linear differential equation. Then the linear combination*

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x), \tag{7}$$

where c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution of the equation.

Teorema 3 (criterion for linearly independent solutions). *Let $y_1(x), y_2(x), \ldots, y_n(x)$ be n solutions of a homogeneous n-th order linear differential equation on an interval I. Then the set of solutions is linearly independent on I iff the Wronskian $W(y_1, y_2, \ldots, y_n) \neq 0$ for every x in I, where*

$$W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}. \tag{8}$$

Teorema 4 (existence of a fundamental set). *There exists a fundamental set of solutions for the homogeneous n-th order linear differential equation on an interval I.*

Teorema 5 (general solution—homogeneous equations). *Let $y_1(x), y_2(x), \ldots, y_n(x)$ be a fundamental set of solutions of a homogeneous n-th order linear differential equation on an interval I. Then the* general solution *of the equation on the interval is given by*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x), \tag{9}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Teorema 6 (general solution–nonhomogeneous equations). *Let $y_p(x)$ be any particular solution of a nonhomogeneous n-th order linear differential equation on an interval I and let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of the associated homogeneous equation on the same interval. Then the* general solution *of the nonhomogeneous equation on the interval is given by*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x), \tag{10}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Teorema 7 (superposition principle—nonhomogeneous equations). *Let $y_{p1}, y_{p2}, \ldots, y_{pk}$ be k particular solutions of a nonhomogeneous n-th order linear differential equation on an interval I corresponding, in turn, to k distinct functions $g_1(x), g_2(x), \ldots, g_k(x)$. That is, suppose y_{pi} denotes a particular solution of the corresponding differential equation*

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g_i(x), \quad i = 1, 2, \ldots, k. \tag{11}$$

then

$$y_p(x) = y_{p1}(x) + y_{p2}(x) + \cdots + y_{pk}(x) \tag{12}$$

is a particular solution of

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g_1(x) + g_2(x) + \cdots + g_k(x). \tag{13}$$

2 Reduction of order

Teorema 8 (reduction of order). *Let $y_1(x)$ be a solution of the homogeneous n-th order linear differential equation on an interval I and that $y_1(x) \neq 0$ for all x in I. Then*

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x)dx}}{y_1^2(x)} dx \tag{14}$$

is a second solution of the equation on the interval.

3 Homogeneous linear equations with constant coefficients

4 Undetermined coefficients - superposition approach

5 Variation of parameters

6 Cauchy-Euler equations

IV The Laplace Transform

1 Definition of the Laplace transform

Definição 1 (Laplace transform). Let f be a function defined on $[0, \infty)$. Then the function F defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)dt \tag{15}$$

is said to be the *Laplace transform* of f. The domain of F is the set of all s for which the improper integral converges.

Teorema 1 (transforms of some basic functions).

(a) $\mathcal{L}\{1\} = \frac{1}{s}$

(b) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$ *for* $n = 0, 1, 2, \ldots$

(c) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

(d) $\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2+\omega^2}$

(e) $\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$

(f) $\mathcal{L}\{\sinh(\omega t)\} = \frac{\omega}{s^2-\omega^2}$

(g) $\mathcal{L}\{\cosh(\omega t)\} = \frac{s}{s^2-\omega^2}$

Definição 2 (exponential order). A function f defined on $[0, \infty)$ is said to be of *exponential order* as $t \to \infty$ if there exist constants c, $M > 0$, and $T > 0$ such that $|f(t)| \leq Me^{ct}$ for all $t > T$. If f is of exponential order, then the Laplace transform $\mathcal{L}\{f(t)\} \leq Me^{ct}$ for all $t > T$.

Teorema 2 (sufficient conditions for existence). *If f is piecewise continuous on $[0,\infty)$ and of exponential order as $t\to\infty$, then the Laplace transform $\mathcal{L}\{f(t)\}$ exists for all $s>c$, where c is a constant such that $|f(t)|\le Me^{ct}$ for all $t>T$.*

Teorema 3 (behavior as $F(s)$ as $s\to\infty$). *If f is piecewise continuous on $[0,\infty)$ and of exponential order as $t\to\infty$ and $\mathcal{L}\{f(t)\}=F(s)$, then $\lim_{s\to\infty}F(s)=0$.*

2 Inverse transforms and transforms of derivatives

Teorema 4 (some inverse transforms). .

(a) $\mathcal{L}^{-1}\left\{\dfrac{1}{s}\right\}=1$

(b) $\mathcal{L}^{-1}\left\{\dfrac{n!}{s^{n+1}}\right\}=t^n$ *for* $n=0,1,2,\ldots$

(c) $\mathcal{L}^{-1}\left\{\dfrac{1}{s-a}\right\}=e^{at}$

(d) $\mathcal{L}^{-1}\left\{\dfrac{\omega}{s^2+\omega^2}\right\}=\sin(\omega t)$

(e) $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2+\omega^2}\right\}=\cos(\omega t)$

(f) $\mathcal{L}^{-1}\left\{\dfrac{\omega}{s^2-\omega^2}\right\}=\sinh(\omega t)$

(g) $\mathcal{L}^{-1}\left\{\dfrac{s}{s^2-\omega^2}\right\}=\cosh(\omega t)$

Teorema 5 (transform of a derivative). *Suppose f, f', ..., f^{n-1} are continuous on $[0,\infty)$, and f, f', ..., f^{n-1} are of exponential order as $t\to\infty$. Then for $s>c$,*

$$\mathcal{L}\{f^{(n)}(t)\}=s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)-\ldots-f^{(n-1)}(0) \tag{16}$$

where $F(s)=\mathcal{L}\{f(t)\}$.

3 Operational properties I

Teorema 6 (first translation theorem). *If $\mathcal{L}\{f(t)\}=F(s)$ exists for $s>c$ and a is any real rumber, then*

$$\mathcal{L}\{e^{at}f(t)\}=F(s-a) \tag{17}$$

for $s>c+a$.

Teorema 7 (second translation theorem). *If $\mathcal{L}\{f(t)\}=F(s)$ exists for $s>c\ge 0$, and a is a positive constant, then*

$$\mathcal{L}\{f(t-a)\,\mathcal{U}(t-a)\}=e^{-as}F(s) \tag{18}$$

for $s>c$.

4 Operational properties II

Teorema 8 (derivatives of transforms). *If the function f is piecewise continuous on $[0,\infty)$, of exponential order and $\mathcal{L}\{f(t)\}=F(s)$, then for $n=1,2,3,\ldots$ and $s>c$,*

$$\mathcal{L}\{t^nf(t)\}=(-1)^n\frac{d^n}{ds^n}F(s). \tag{19}$$

Teorema 9 (convolution theorem). *If the functions f and g are piecewise continuous on $[0,\infty)$, of exponential order and $\mathcal{L}\{f(t)\}=F(s)$, $\mathcal{L}\{g(t)\}=G(s)$, then for $s>c$,*

$$\mathcal{L}\{f*g\}=\mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}=F(s)G(s). \tag{20}$$

Teorema 10 (transform of a periodic function). *If f is piecewise continuous $[0, \infty)$, of exponential order, and periodic with period T, then*

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt. \tag{21}$$

5 The Dirac-Delta function

Teorema 11 (transform of the dirac delta function). *For $t_0 > 0$,*

$$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}. \tag{22}$$

V Systems of Linear Differential Equations

1 Theory of linear systems

Definição 1 (solution vector). A *solution vector* on an interval I is any column matrix

$$\mathbf{X} = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} \tag{23}$$

whose entries are differentiable functions satisfying the system $\mathbf{X}' = \mathbf{AX} + \mathbf{F}$ on the interval.

Teorema 1 (existence of a unique solution). *Let the entries of the matrices $\mathbf{A}(t)$ and $\mathbf{F}(t)$ be functions continuous on a common interval I that contains the point t_0. Then there exists a unique solution of the initial-value problem $\mathbf{X}' = \mathbf{A}(t)\mathbf{X} + \mathbf{F}(t)$, $\mathbf{X}(t_0) = \mathbf{X_0}$ on the interval.*

Teorema 2 (superposition principle). *Let $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_k$ be a set of solution vectors of the nonhomogeneous system $\mathbf{X}' = \mathbf{AX} + \mathbf{F}$ on an interval I. Then the linear combination*

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_k\mathbf{X}_k, \tag{24}$$

where c_i, $i = 1, 2, \ldots, k$ are arbitrary constants, is also a solution of the system on the interval.

Teorema 3 (criterion for linearly independent solutions). *Let $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ be n solution vectors of a homogeneous system on an interval I. Then the set of solution vectors is linearly independent on I iff the Wronskian*

$$W(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n) = \begin{vmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{vmatrix} \neq 0 \tag{25}$$

for every t in the interval.

Teorema 4 (existence of a fundamental set). *There exists a fundamental set of solutions for the homogeneous system $\mathbf{X}' = \mathbf{AX}$ on an interval I.*

Teorema 5 (general solution–homogeneous systems). *Let $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ be a fundamental set of solutions of a homogeneous system on an interval I. Then the* general solution *of the system on the interval is given by*

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n, \tag{26}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

Teorema 6 (general solution–nonhomogeneous systems). *Let $\mathbf{X}_p$ be any particular solution of a nonhomogeneous system on an interval I and let $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ be a fundamental set of solutions of the associated homogeneous system on the same interval. Then the* general solution *of the nonhomogeneous system on the interval is given by*

$$\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n + \mathbf{X}_p, \tag{27}$$

where the c_i, $i = 1, 2, \ldots, n$ are arbitrary constants.

2 Homogeneous linear systems

Teorema 7 (distinct real eigenvalues: general solution—homogeneous systems). *Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be n distinct real eigenvalues of the coefficient matrix $\mathbf{A}$ of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ and let $\mathbf{K}_1, \mathbf{K}_2, \ldots, \mathbf{K}_n$ be the corresponding eigenvectors. Then the* ***general solution*** *of the system on the interval $(-\infty, \infty)$ is given by*

$$\mathbf{X} = c_1\mathbf{K}_1e^{\lambda_1 t} + c_2\mathbf{K}_2e^{\lambda_2 t} + \cdots + c_n\mathbf{K}_ne^{\lambda_n t}. \tag{28}$$

Teorema 8 (solutions corresponding to a repeated eigenvalue). *Let λ_1 be a real eigenvalue of multiplicity k of the coefficient matrix $\mathbf{A}$ of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ and let $\mathbf{K}_1$ be an eigenvector corresponding to λ_1. Then the k linearly independent solutions of the system corresponding to the eigenvalue λ_1 are given by*

$$\mathbf{X}_1 = \mathbf{K}_1e^{\lambda_1 t} \tag{29}$$

$$\mathbf{X}_2 = t\mathbf{K}_1e^{\lambda_1 t} \tag{30}$$

$$\vdots \tag{31}$$

$$\mathbf{X}_k = t^{k-1}\mathbf{K}_1e^{\lambda_1 t} \tag{32}$$

Teorema 9 (real solutions corresponding to complex eigenvalues). *Let $\lambda_1 = \alpha + i\beta$ be a complex eigenvalue of the coefficient matrix $\mathbf{A}$ of the homogeneous system $\mathbf{X}' = \mathbf{AX}$ and let $\mathbf{K}_1$ be a corresponding eigenvector. Then the two linearly independent solutions of the system on the interval $(-\infty, \infty)$ are given by*

$$\mathbf{X}_1 = Re\left\{\mathbf{K}_1e^{\lambda_1 t}\right\} = e^{\alpha t}\left[Re\{\mathbf{K}_1\}\cos(\beta t) - Im\{\mathbf{K}_1\}\sin(\beta t)\right] \tag{33}$$

$$\mathbf{X}_2 = Im\left\{\mathbf{K}_1e^{\lambda_1 t}\right\} = e^{\alpha t}\left[Re\{\mathbf{K}_1\}\sin(\beta t) + Im\{\mathbf{K}_1\}\cos(\beta t)\right] \tag{34}$$

3 Nonhomogeneous linear systems

VI Dynamics of First-Order Difference Equations

The following is an introduction to difference equations, based on *An Introduction to Difference Equations* (3e), by S. Elaydi.

Definição 1. A linear *homogeneous* first-order difference equation is given by

$$x(n+1) = a(n)x(n), \quad x(n_0) = x_0, \quad n \geq n_0 \geq 0, \tag{35}$$

and the associated *nonhomogeneous* equation is

$$y(n+1) = a(n)y(n) + g(n), \quad y(n_0) = y_0, \quad n \geq n_0 \geq 0, \tag{36}$$

where in both cases $a(n) \neq 0$, and $a(n)$ and $g(n)$ are real-valued functions defined for $n \geq n_0$.

Proposição 1. *The solution to* (35) *can be obtained by iteration and mathematical induction and is given by*

$$x(n) = \left[\prod_{i=n_0}^{n-1} a(i)\right] x_0 \tag{37}$$

For the nonhomogeneous case, the unique solution is

$$y(n) = \left[\prod_{i=n_0}^{n-1} a(i)\right] y_0 + \sum_{r=n_0}^{n-1} \left[\prod_{i=r+1}^{n-1} a(i)\right] g(r). \tag{38}$$

Definição 2 (equilibrium point). A point x^* in the domain of f is said to be an *equilibrium point* of a difference equation

$$x(n+1) = f(x(n)) \tag{39}$$

if it is a fixed point of f, i.e., $f(x^*) = x^*$.

Definição 3 (eventually equilibrium point). Let x be a point in the domain of f. If there exists a positive integer r and an equilibrium point x^* of (39) such that $f^r(x) = x^*$, $f^{r-1} \neq x^*$, then x is an *eventually equilibrium (fixed) point.*

Definição 4. (a) The equilibrium point x^* of (39) is *stable* if given $\epsilon > 0$ there exists $\delta > 0$ such that $|x_0 - x^*| < \delta$ implies $|f^n(x_0) - x^*| < \epsilon$ for all $n > 0$. If x^* is not stable, then it is called *unstable.*

(b) The point x^* is said to be *attracting* if there exists $\eta > 0$ such that

$$|x(0) - x^*| < \eta \implies \lim_{n\to\infty} x(n) = x^*.$$

If $\eta = \infty$, x^* is called a *global attractor* or *globally attracting.*

(c) The point x^* is an *asymptotically stable equilibrium point* if it is stable and attracting. If $\eta = \infty$, x^* is said to be *globally asymptotically stable.*